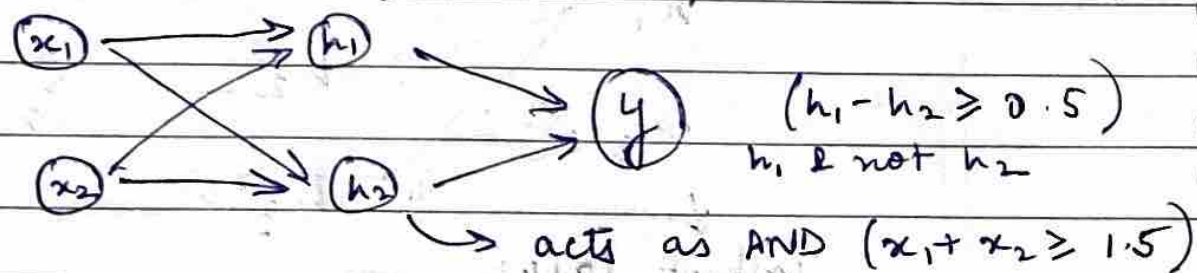
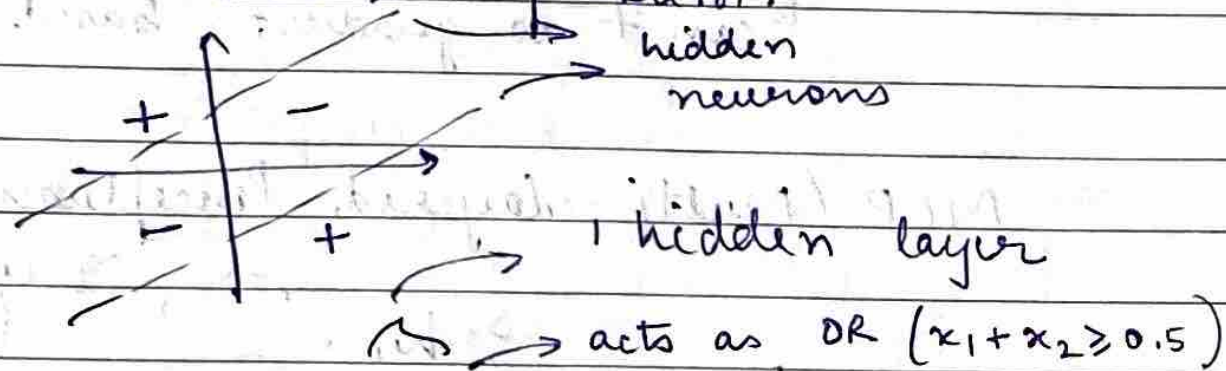


27th Jan 2026 / DA6401

- we were discussing how the hidden layers of MLP solved the XOR problem.



x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

XOR Truth Table

- _/_/_
- basically, MLP helps to transform the input data pts. into a feature space st. the problem becomes linearly separable.

$L1 \rightarrow m$ neurons, $L2 \rightarrow n$ neurons

we have $m \times n$ connections, wts are a vector (w)

$$h = \underbrace{\sum (w_i x_i + b)}_{\text{sum}} \xrightarrow{\text{heuristic}}$$

neuron $i \rightarrow$ wts. for neuron i in the hidden layer. It detects 1 specific feature.

$$x \in \mathbb{R}^3 \rightarrow h \in \mathbb{R}^2$$

$$\Rightarrow W = 2 \times 3$$

- use many neurons, slowly we can approx. any funcⁿ as sum of many linear neurons. (or any mathematical funcⁿ in the universe)

$\rightarrow$ (universal approx. th. $\rightarrow$ MLPs can app. any conti. funcⁿ.)

- Next: UAT, Modern Neuron, Activation Func^{ns} (Sigmoid, Tanh, ReLU)

